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Q1) Define Business Intelligence. Explain it's key components with road diagram:

Business Intelligence

Business Intelligence is a set of process, architectures and ^{tech} ~~the~~ ^{tools} ~~the~~ that convert raw data into meaningful and useful information to support better business decision making.

It helps organization analyze historical and current data identify trends and patterns and improve strategic, tactical and operational decision.

Key components of Business Intelligence.

Data Source:

Data is collected from operational databases ERP/CRM systems, files and external sources.

ETL (Extract, Transform, Load).

Extract data from sources, clean and transforms it, and loads it into the data warehouse.

Data warehouse:

A centralized repository that stores integrated and historical data for analysis.

OLAP & BI Tools:

provide multi dimensional analysis, reports, dashboards and data visualization.

End Users:

Managers and decision makers use BI outputs to make informed business decisions.

Data Sources
(ERP, CRM, Files)

ETL
Extract transform Load

Data warehouse

OLAP & BI Tools
Reports Dashboard
Analytics

End user
decision makers

Q2) Explain the difference between data, information and knowledge with suitable example:

Data:

Data are raw figures and facts collected to be processed. They can be numbers, words, images or sounds and are typically ^{un}processed. Data may be linked to be the raw material of information.

eg:

Temperature reading (eg. 58°F , 22°C),
customer satisfaction ratings (eg. 4 out of 10),
web traffic statistics (eg. 1500 daily visits).

Information:

Information is the output of processed, organised or structured data presented in a form that can be

understood. By taking temperature readings, for example, and reporting them as trends over time, you transform raw data into meaningful information. Information answers questions such as who, what, where and when.

eg:

- A report showing that the average temperature for the ^{week} next was 69.8°F (21°C), indicating it was a warmer-than-average week.
- A sales report indicating a 15% increase in sales last quarter compared to the previous quarter.

knowledge:

knowledge is the realization and understanding gained through experience, learning and the application of information. It's the ability to read and apply information in a specific context.

eg:

- A gardener's understanding that plants are damaged when temperatures drop below 59°F (15°C) enabling them to take appropriate action to protect the plants.
- A marketing manager using feedback from customer surveys to design an effective targeted marketing campaign.
- A scientist applying the laws of physics to solve complex problem by ^{selecting} the most efficient experimental design.

Q3) Describe the advantages and applications of Business Intelligence in organizations.

Advantages of Business Intelligence:

- Improved Decision making.
BI provides accurate and timely information, helping managers make informed decisions.
- Better Business performance.
Helps in identifying trends, opportunities and problem areas to improve efficiency.
- Enhanced Data Accuracy.
Integrates data from multiple sources and ensures data consistency.
- Time Saving.
Automated reports and dashboards reduce manual work and save time.
- Competitive advantage.
Organizations can respond quickly to market changes and customer needs.

Applications of Business Intelligence:

- Sales and Marketing.
Analyzes sales trends, customer behaviour, and campaign performance.
- Finance.
Used for budgeting, forecasting, profit analysis and financial reporting.
- Human Resources.
Helps in employee performance analysis, attrition analysis and workforce planning.

supply chain management.
Improved inventory control, demand forecasting and supplier performance.

- customer Relationship Management (CRM).
Enhance customer satisfaction through personalized service and retention analysis.

Q4) Explain how BI Supports effective and timely decision making.

Business Intelligence (BI) plays a vital role in helping organizations make accurate, fast and informed decisions by transforming raw data into meaningful insights.

Real time and accurate data.

BI system provides access to current and reliable data from multiple sources, reducing guesswork and errors in decisions.

Data integration.

BI integrates data from different department such as sales, finance and HR, giving a complete and unified view of business operations.

Analytical Capabilities.

Using tools like reports, dashboards and OLAP BI enables trend analysis, comparisons and forecasting which support strategic planning.

Faster Reporting.

Automated reporting and visualization tools reduce manual effort and deliver timely information to decision makers.

Improved strategic and operational decisions

It helps management opportunities detect problem early and take corrective actions at the right time.

Qn) Explain how data will be converted into information and knowledge.

In Business Intelligence, raw data is transformed step-by-step into information and then into knowledge to support decision making.

Data Collection:

Raw data is collected from various sources such as databases, transactions, sensors and business applications.

At this stage data has no meaning or context.

ex: 100, 150, 120.

Data processing:

Data is cleaned, filtered, sorted and organized using ETL process to remove errors and inconsistencies.

ex: Sales amount arranged customer wise and date wise.

Information generation:

processed data is summarized and presented in meaningful form such as reports, charts or tables. This provide context and meaning.

ex: Total Sales for the day is ₹ 450.

Analysis and Interpretation:

Information is analysed using BI tools (Col Ap, ~~database~~ ^{dashboard}, analytics) to identify patterns and trends.

ex: Sales are higher during weekends.

Knowledge creation:

Based on the analysis and experience, insights are derived that help in decision making.

This stage represents knowledge.

ex: offering weekend discount can increase overall sales.